

CSCE 650 VIRTUAL REALITY · MODULE 11 · LECTURE NOTES

Spatial Audio

The half of presence that gets a tenth of the effort.

MODULE OUTCOMES

- Explain how humans localise sound.
- Explain HRTFs and why they are personal.
- Explain the role of room acoustics in distance perception.
- Apply the audio techniques that most improve presence.
- Explain why audio is a disproportionate return on effort.

1 How humans localise sound

Cue	Mechanism	What it resolves
Interaural time difference (ITD)	Sound from the left reaches the left ear fractionally earlier — up to about 0.7 ms.	Left and right , for frequencies below roughly 1.5 kHz, where the wavelength exceeds the head's width and phase is unambiguous.
Interaural level difference (ILD)	The head shadows the far ear, reducing the level there.	Left and right , for frequencies above roughly 1.5 kHz, where the head is large enough relative to the wavelength to cast an acoustic shadow.
Spectral cues (the HRTF)	Direction-dependent filtering by the pinna, head, and torso.	Up and down, front and back. §1.1.
Head movement	How the above cues change as the head turns.	Resolves ambiguity — a sound in front and one behind produce opposite changes when you turn, so a small movement disambiguates them immediately.
Reverberation	The ratio of direct to reflected energy.	Distance , and the size and material of the space. §2.

ITD and ILD alone leave a 'cone of confusion'. For any given pair of time and level differences there is a whole cone of directions producing them — front and back, up and down are indistinguishable. **This is why naive stereo panning sounds like it is inside your head** rather than out in the world: the cues are present but incomplete, and the brain declines to place the sound externally.

THE PINNA IS WHY YOU CAN TELL UP FROM DOWN

Sound arriving from different directions reflects and diffracts differently off the folds of the outer ear, the head, and the shoulders. The pinna's convolutions are not decorative; they are a direction-encoding filter.

This imposes a direction-dependent frequency response — certain frequencies reinforced, others notched out, with the pattern varying continuously with direction. A sound from above has a different spectral signature from the same sound in front.

The brain has learned your particular filter over a lifetime of hearing things whose positions it could verify, and uses it to resolve elevation and front/back — the dimensions ITD and ILD cannot reach.

A head-related transfer function is a measurement of that filter, as a function of direction. Convolving a monophonic sound with the appropriate HRTF for a given direction places it convincingly in space over headphones, which is the whole basis of VR audio.

HRTFS ARE PERSONAL, AND GENERIC ONES ARE USED ANYWAY

Your HRTF is determined by the shape of your ears, head, and shoulders, which vary as much between people as fingerprints do. Pinna shape in particular is highly individual.

Measuring one properly requires an anechoic chamber, a speaker array or a moving source, microphones in the ear canals, and a few hours of the subject's time. This is not going to happen for your users.

So generic HRTFs are used — measured from a dummy head, or averaged across a population, with some systems offering a small set to choose between or fitting one from a photograph of the ear. They work reasonably well for most listeners.

The characteristic failure is elevation error and front/back confusion: a sound that should be in front is perceived behind, or a sound overhead is heard at ear level. **Head movement largely rescues it** — the cues change in opposite ways for front and back, so a small turn resolves the ambiguity. **Which is precisely why VR audio works better than static binaural recordings:** the listener moves, continuously, and every movement disambiguates.

2 Room acoustics

REVERBERATION IS HOW YOU HEAR DISTANCE AND SIZE

The direct sound tells you the direction. The reflections tell you the room — how large it is, what it is made of, and how far away the source is.

The direct-to-reverberant ratio is the primary distance cue. A nearby source is dominated by its direct sound; a distant one is dominated by reverberation, because the direct path has attenuated while the reflected energy in the room has not. **Loudness alone is a poor distance cue** — a quiet sound nearby and a loud sound far away have the same level — and this ratio is what disambiguates them.

Early reflections, within roughly the first 80 ms, convey geometry. The first few bounces arrive from specific directions with specific delays, and they encode the size of the space and the materials of its surfaces.

Late reverberation conveys overall character — the decaying diffuse tail. Together, early and late reverberation are why a cathedral and a cupboard are unmistakably different with your eyes closed, before anyone has said a word.

- **Per-source HRTF spatialisation of the direct path.** The baseline, and every modern audio engine provides it. Without it nothing else matters.
- **Early reflections computed from the actual geometry** — by the image-source method for simple rooms, or by ray tracing for complex ones. **This is what makes a room sound like *that* room** rather than like a generic room-shaped reverb, and it is the step that most distinguishes convincing VR audio from adequate VR audio.
- **A reverb matched to the space**, varying between rooms, with smooth interpolation as the listener moves between zones. An abrupt reverb change at a doorway is as jarring as a visual pop.
- **Occlusion and obstruction.** A sound behind a wall should be *muffled*, not merely quieter — walls attenuate high frequencies far more than low ones. **A simple low-pass filter is most of the effect**, it is cheap, and it is the single most noticeable improvement available after basic spatialisation.
- **Distance attenuation with air absorption.** Distant sounds lose high frequencies as well as level, which is why a far-off voice sounds dull as well as quiet.

3 Why audio matters disproportionately

AUDIO COVERS THE WHOLE SPHERE; VISION COVERS 110 DEGREES

The user sees roughly 110 degrees and hears everything. Audio is the only sense in a headset that covers the full sphere, continuously, without the user having to orient toward anything.

So audio is how a space feels inhabited. Things behind you, above you, around a corner, and out of sight exist because you can hear them. A visually detailed room with no sound behind the listener feels like a diorama; the same room with footsteps and distant activity feels like a place.

And it directs attention without taking control — which Module 01 identified as the central design problem of a medium where you cannot cut or frame. A sound from the left makes people look left, voluntarily and immediately, and they experience it as their own choice.

It is therefore the primary tool for guiding a user through an experience, and the one that most reliably works without breaking presence.

- **Measurably increased presence.** Studies consistently find that adding spatial audio raises presence ratings more than comparable investment in visual fidelity does.
- **Perceived visual quality rises.** Good audio makes people rate the *graphics* more highly — a robust, well-replicated, and slightly absurd finding that is worth knowing when arguing for audio budget.
- **Reduced sickness in some studies**, plausibly because a stable, world-locked sound field acts as a reference frame consistent with the vestibular signal (Module 07).
- **Better spatial understanding.** Users build more accurate mental maps of environments with good spatial audio.
- **And all of it for a small fraction of the engineering cost of the graphics. Audio is the highest return per unit of effort available in VR**, and it is routinely the last thing anyone does, assigned to whoever is free in the final week.

AUDIO MUST BE HEAD-TRACKED, WITH LOW LATENCY

When the head turns, the sound field must remain fixed in the world — exactly as the visual scene does. A sound source to the user's left must move to in front of them when they turn left.

If the sound field rotates with the head, every source is locked to the listener, which destroys the spatial illusion completely. It is immediately obvious once you know to listen for it, and it is a common bug in applications where the audio engine was not informed of the head pose.

The audio latency budget is tighter than people assume. Head-tracked audio lag becomes perceptible above roughly 30 ms, and **audio buffers are frequently sized generously by default** because flat applications have no reason to care — so this is a setting that must be checked rather than assumed.

And audio-visual synchrony matters. A sound arriving noticeably after its visible cause breaks the perceptual association between them, and the object stops being the thing that made the sound.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Meta — Audio design and spatialisation documentation (free)	Practical VR audio: HRTF spatialisation, reverb zones, occlusion, and the latency requirements of §3.
Steam Audio documentation (free, with SDK)	Geometry-based early reflections and occlusion — §2's implementation, free and usable in both major engines.
Begault — 3-D Sound for Virtual Reality and Multimedia (NASA, free)	The foundational text on binaural audio and HRTFs, free from NASA. Thorough on §1.
SOFA HRTF databases (free)	Measured HRTF sets in a standard format, including several public databases. Useful for experimenting with individual variation.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Listen to a sound spatialised by simple stereo panning and by HRTF convolution. Describe where each appears to be.
- 2 Find your own front/back confusion: have a spatialised sound placed directly in front and directly behind, without visuals, and try to distinguish them. Then allow head movement and repeat.
- 3 Try three different generic HRTFs from a public database and report which localises best for you.
- 4 Implement direct-to-reverberant ratio as a distance cue and verify that listeners can judge distance without level changes.
- 5 Implement early reflections by the image-source method for a simple rectangular room and compare against a generic reverb.
- 6 Add a low-pass occlusion filter for sounds behind walls and measure how much it improves the sense of the space.
- 7 **Break head tracking in the audio** — rotate the sound field with the head — and confirm how obvious it is.
- 8 Measure your audio pipeline's latency and compare against the 30 ms threshold.
- 9 Show the same VR scene to six people with and without spatial audio, and ask them to rate the *graphics*. Report whether the finding in §3 replicates.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name five localisation cues and say what each resolves.
- 2 What is the cone of confusion, and why does stereo panning sound internal?
- 3 What does the pinna do, and what is an HRTF?
- 4 Why are generic HRTFs acceptable, and what rescues their failure mode?
- 5 What is the primary distance cue, and why is loudness insufficient?
- 6 Distinguish early reflections from late reverberation.
- 7 Give five elements of a convincing spatial audio implementation.
- 8 Why does audio matter disproportionately in VR? Give the structural argument.
- 9 Name four measured benefits of good VR audio.
- 10 What must be true of head tracking in audio, and what is the latency threshold?